

Group - 5

Lab Report

Experiment # 4 : Design and Implementation
of 4-bit Parallel Binary
Adder

Experiment # 5 : Implementation of 4-bit
Magnitude Comparator

Experiment # 6 : Design circuits using encoder
and decoder

Experiment # 7 : Function Implementation using
MUX

Required Components for Lab-4

- (1) IC 7408
- (2) IC 7432
- (3) IC 7486
- (4) IC 7483

Required Components for Lab-5

(1) IC 7408

(2) IC 7432

(3) IC 7404

(4) IC 4077

Required Components for Lab-6

(1) IC 74138

(2) IC 74148

Required Components for Lab-7

(1) IC 74153

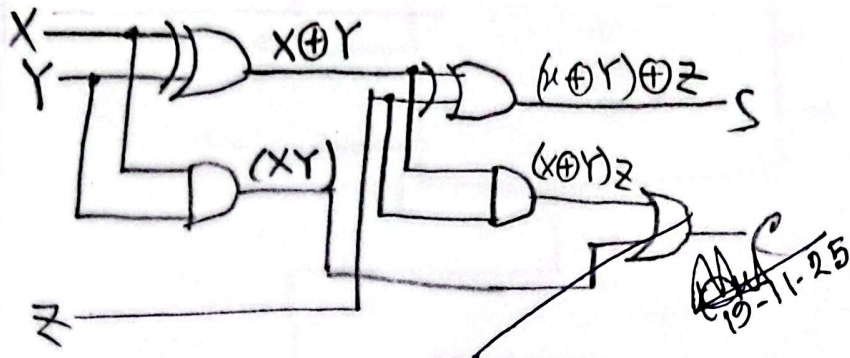
(2) IC 7408

(3) IC 7432

(4) IC 7404

Experiment 4

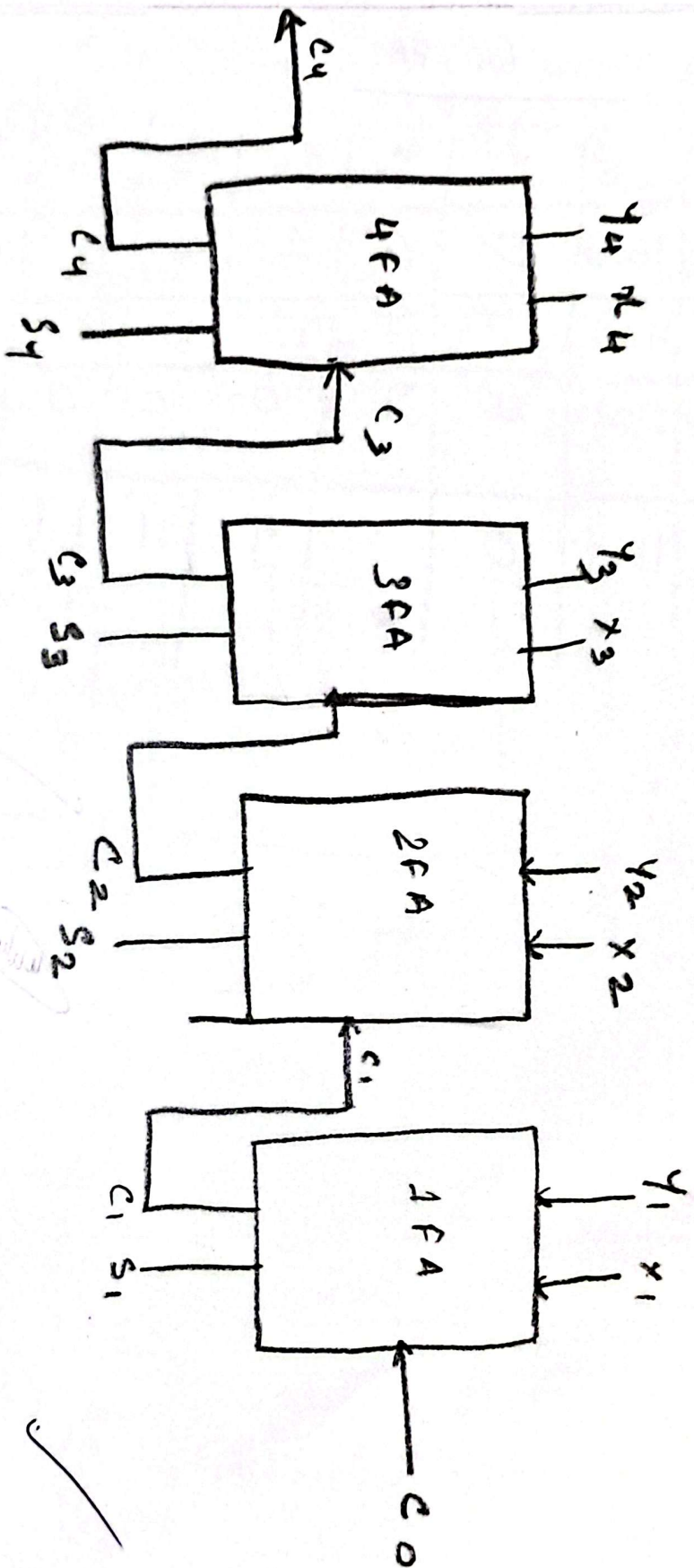
Full Adder Circuit diagram:



$$C = XY + (X \oplus Y)Z$$

$$S = (X \oplus Y) \oplus Z$$

Block Diagram for FA



Truth Table for PA

A	B	c _{in}	c ₄	s ₄	s ₃	s ₂	s ₁
0000	1000	0	0	1	0	0	0
1100	1101	0	1	1	0	0	1
1110	0011	0	1	0	0	0	1
1111	1111	0	1	1	1	1	0



worky

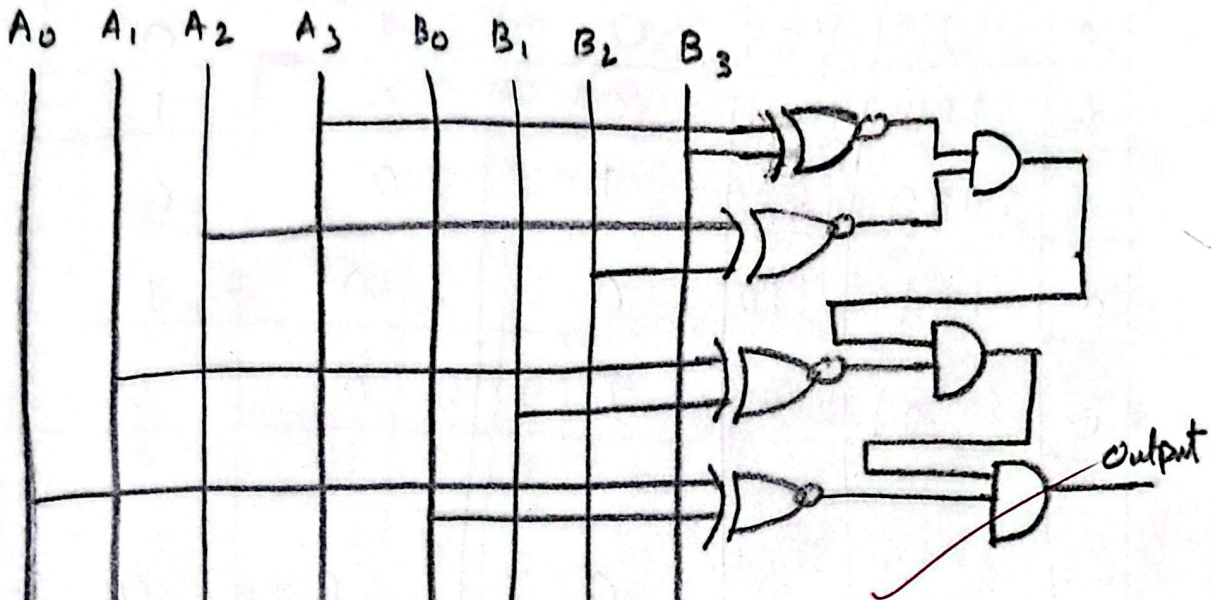
EXP-05:

Truth Table:

	A	B	$A = B$	$A > B$	$A < B$
a.	1011	0101	0	1	0
b.	0100	1001	0	0	1
c.	1000	1000	1	0	0
d.	1101	1110	0	0	1
e.	1001	1001	1	0	0
f.	0010	0001	0	1	0

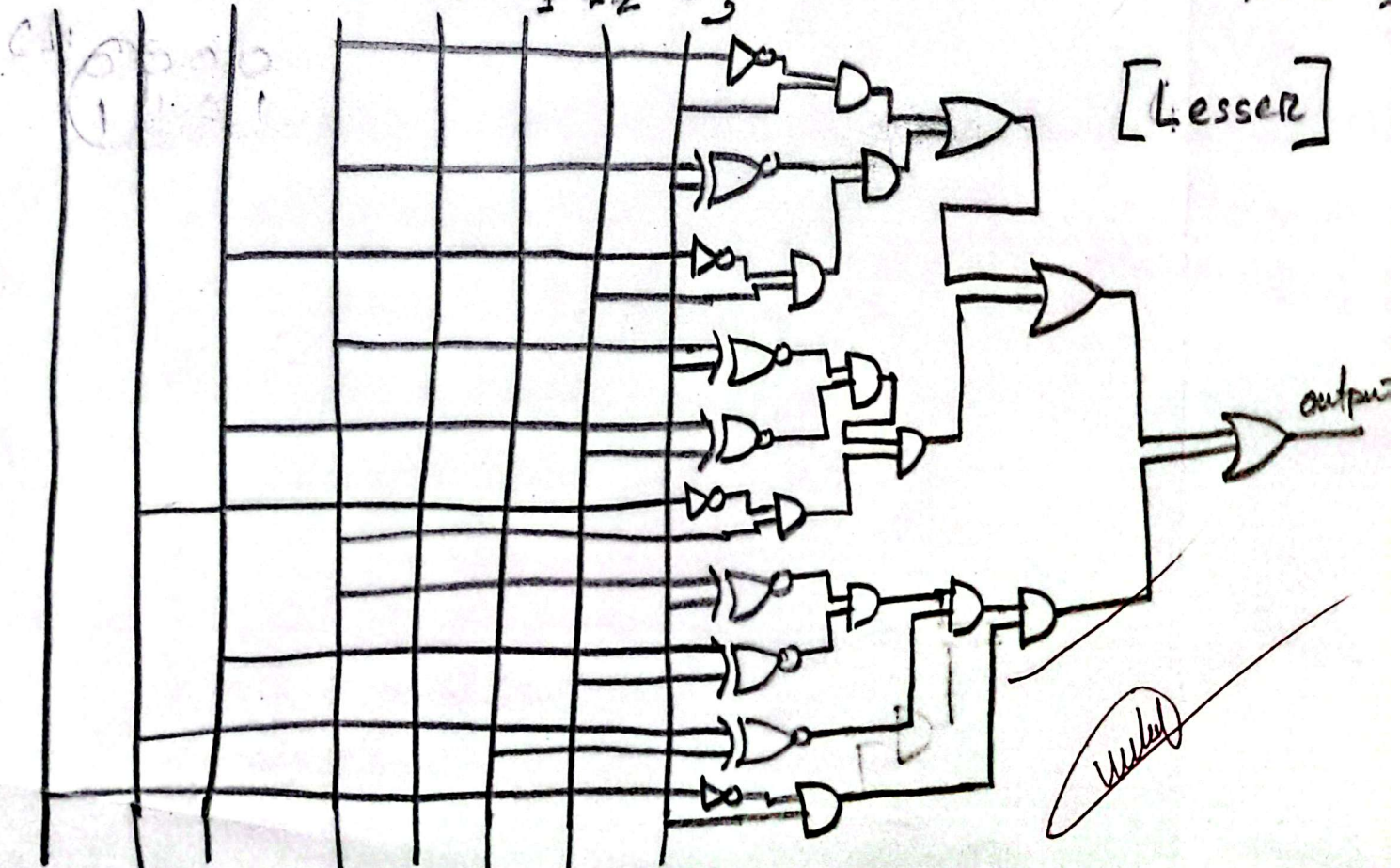
[Equal]

$$(A_3 \odot B_3) \cdot (A_2 \odot B_2) \cdot (A_1 \odot B_1) \cdot (A_0 \odot B_0)$$



$$\bar{A}_3 B_3 + (A_3 \odot B_3)(\bar{A}_2 \cdot B_2) + (A_3 \odot B_3)(A_2 \odot B_2)(\bar{A}_1 \cdot B_1) + (A_3 \odot B_3)(A_2 \odot B_2)(A_1 \odot B_1)(\bar{A}_0 \cdot B_0)$$

[Lesser]

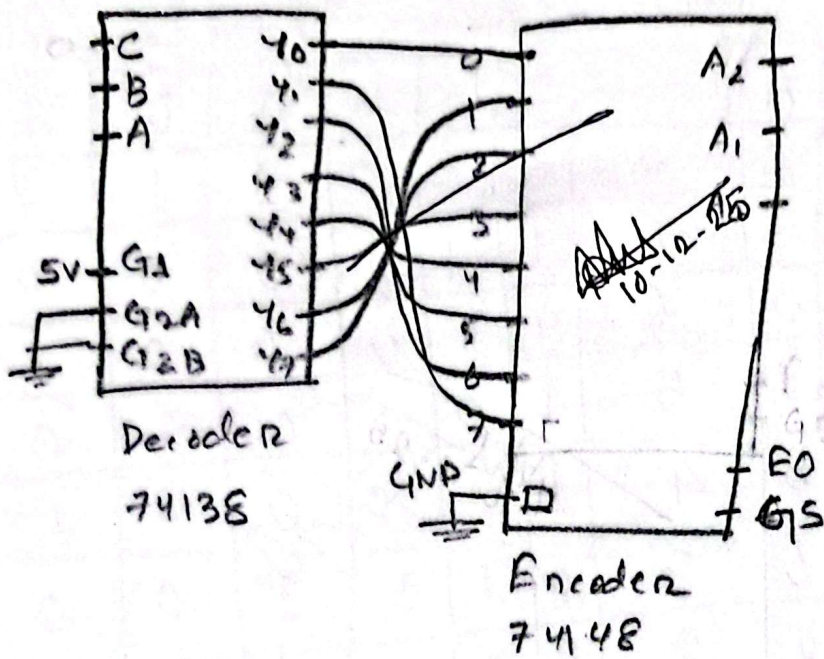


EXP 06: Design Circuit using encoders & Decoders

Truth Table:

Inputs				Expected Outputs				Active Low Outputs		
Minterm	C	B	A	Minterm	D ₂	D ₁	D ₀	D ₂	D ₁	D ₀
0	0	0	0	0	0	0	0	1	1	1
1	0	0	1	7	1	1	1	0	0	0
2	0	1	0	6	1	1	0	0	0	1
3	0	1	1	5	1	0	1	0	1	0
4	1	0	0	4	1	0	1	0	1	1
5	1	0	1	3	0	1	1	1	0	0
6	1	1	0	2	0	1	0	1	0	1
7	1	1	1	1	0	0	1	1	1	0

Block Diagram



	T_0	T_1	T_2	T_3
$A'B'$	0	1	2	3
$A'B$	4	5	6	7
AB'	8	9	10	11
AB	12	13	14	15
	B	$\bar{B}$	A	$\bar{A}$

$$A'B' + A'B + AB'$$

$$A'(B + B') + AB'$$

$$A' + AB'$$

~~1-12-25~~

$$(A' + A)(A' + B')$$

$$A' + B'$$

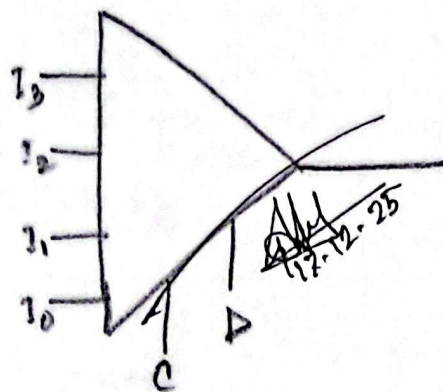
$$T_0 = B$$

$$T_1 = \bar{B}$$

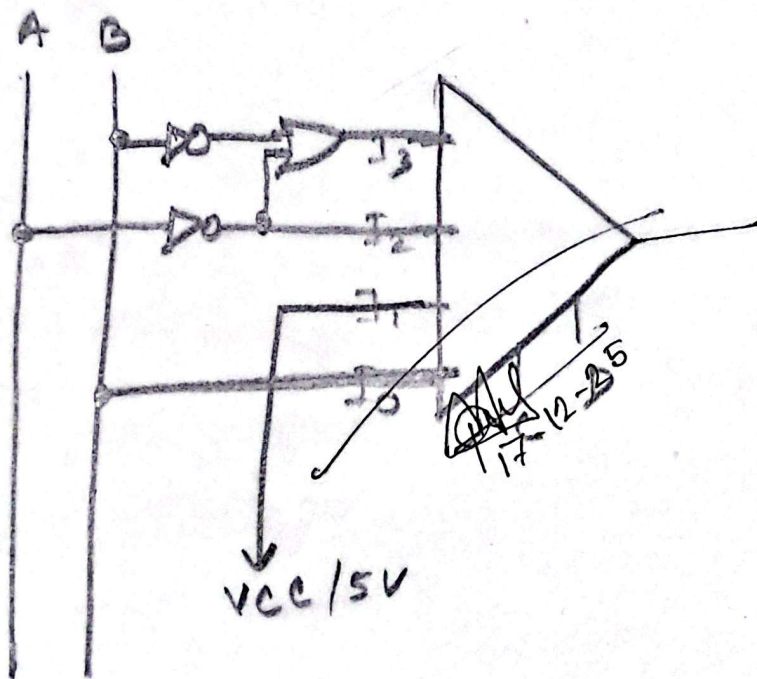
$$T_2 = A$$

$$T_3 = A + B$$

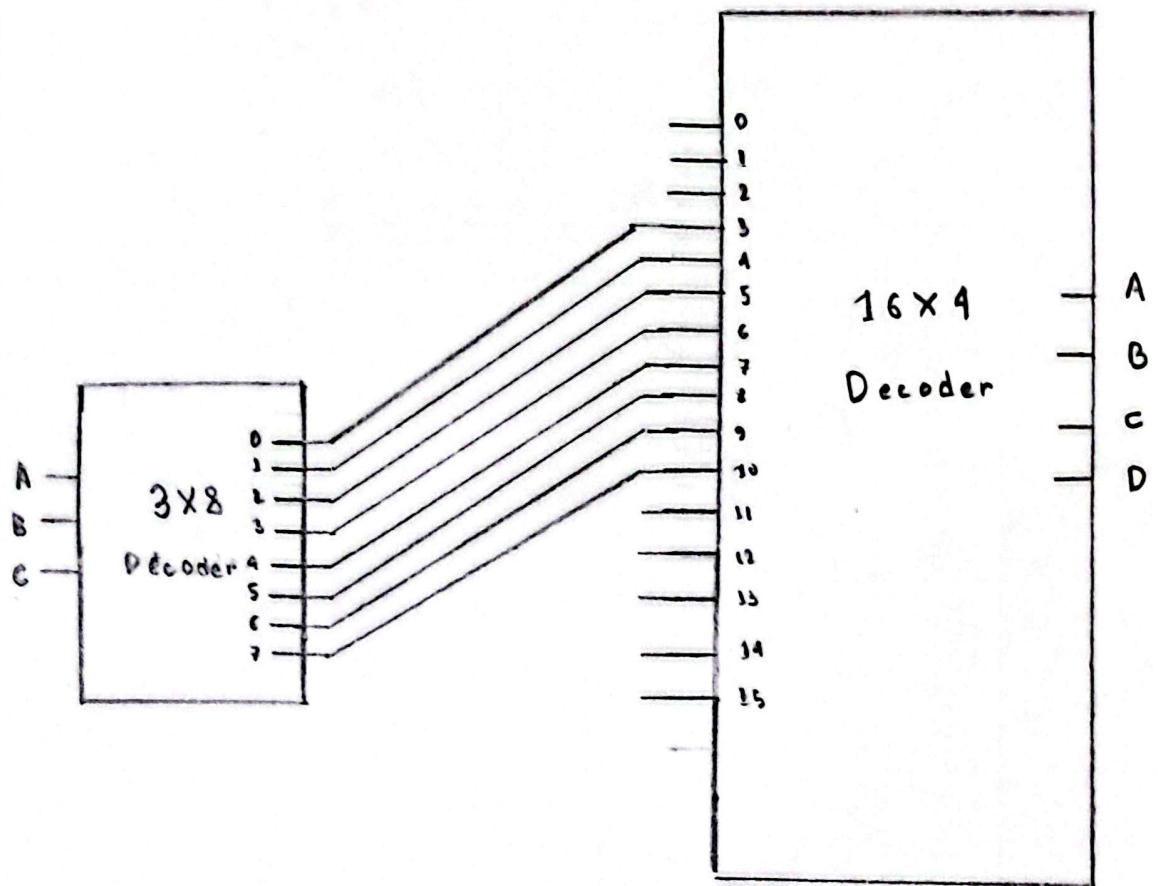
Experiment No. 7



Ckt Diagram for $F(A, B, C) = \sum (1, 2, 3, 4, 5, 6, 7, 9, 11, 12, 13)$



Answer to the Question No. 1



Answer to the Question No. 2

2 bit Parallel Adder using 8:1 MUX's

